microstructure-lab

An event-driven framework for market-microstructure simulation and execution analytics

Mengren Man • April 2026

Summary. *microstructure-lab* is a Python research repository that exercises the full hypothesis → simulation → diagnostics → iteration loop on short-horizon microstructure and crypto-carry ideas. Every friction that typically breaks naive backtests – bid–ask spread, stochastic fill probability, adverse selection, inventory accumulation, fees – is a first-class, measurable quantity. The repo ships four strategies, a ten-metric analytics suite, four research notebooks, a 24-test suite, and GitHub Actions CI across Python 3.10–3.12. All results below are on synthetic data and are intended as diagnostics, not return claims.

1. Framework

Simulation engine. Event-driven limit-order book with multi-level depth (`set_depth`, `depth_imbalance`, `weighted_mid`), stochastic passive fill (exponential-decay in distance-to-mid), a vol-scaled aggressive-cross model, fee-aware mark-to-mid PnL, and per-fill adverse-selection annotation (mid before/after a configurable horizon). Runs are deterministic under a seed and fully config-driven.

Market scenario. Geometric Brownian mid with Markov regime switching (calm/stressed) and vol-dependent spread widening, producing the state-contingent flow the microstructure literature actually cares about.

Strategies (swap via JSON `"type"` field; no code change):

Strategy	Role	Signal
`InventorySkewMM`	Passive maker	Inventory skew
`TWAPStrategy`	Passive taker	Time-sliced
`MomentumStrategy`	Aggressive taker	Rolling return
`DoNothingStrategy`	Control baseline	—

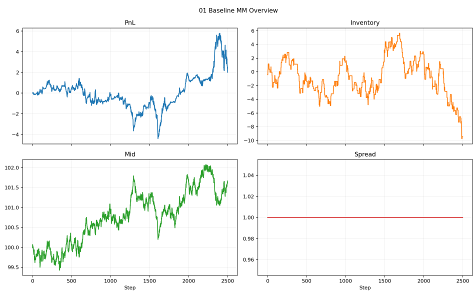

Figure 1: Representative `InventorySkewMM` trajectory (2,500 steps): mark-to-mid PnL, inventory, mid-price, and quoted spread.

Analytics (per run). `final_pnl`, `max_drawdown`, `sharpe_annualized`, `fills`, `fees_paid`, `avg_abs_inventory`, `inventory_half_life`, `fill_rate`, `realized_spread_avg`, `adverse_selection_avg`.

2. Findings across the four notebooks

(1) Baseline market maker. With the calibrated inventory penalty λ=7.5 bps (`configs/baseline_mm.json`, σ=4 bps, 2,500 steps), the maker delivers `final_pnl`= +1.72, annualised Sharpe +2.31, 377 fills, and a *net fee rebate* of −1.17. Inventory mean-reverts with a half-life of 118 steps vs. ∞ when $\lambda \lesssim 6$ bps; above $\sim$8 bps the skew starts crossing the spread and compresses realised spread. NB01 sweeps the full λ grid so the trade-off is visible rather than asserted.

(2) Execution quality. NB02 decomposes the fill stream: 100% maker fills for the baseline (expected – it never crosses), a net maker rebate, and empirical distributions for a slippage proxy and a post-fill adverse-selection proxy. The adverse-selection histogram is the diagnostic that matters: its mean is the portion of the quoted spread being given back to informed flow (Glosten–Milgrom), and it is the lever that separates real edge from apparent edge.

(3) Funding / basis carry. NB03 models a perp–spot basis as an OU process ($\sim$8% annualised on the synthetic series) and runs a delta-neutral carry strategy with a threshold-based entry rule. The equity curve accrues monotonically as designed, and a threshold sweep quantifies the turnover/fee trade-off. This is the template for plugging in real Binance perpetual-funding history (see roadmap).

(4) Two-venue latency arbitrage. NB04 sweeps signal latency × transfer delay on a cross-venue basis signal (Figure 2). The edge collapses the moment signal latency rises above zero – a canonical result that motivates co-location spend, and one any live cross-venue build must respect before scaling.

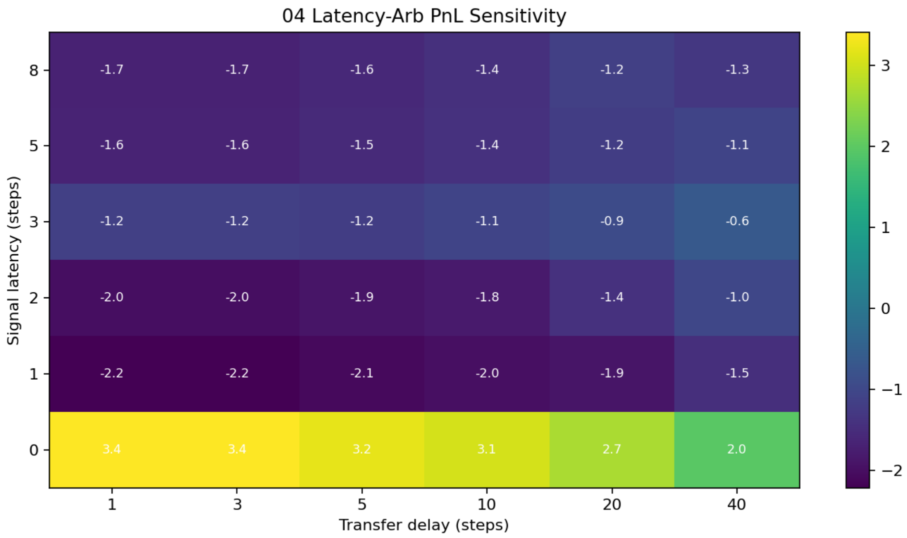

Figure 2: Latency-arbitrage PnL sensitivity: only the zero-signal-latency row survives. Transfer delay matters materially less than information latency over this grid.

Calibrated baseline snapshot

`baseline_mm.json` • σ=4 bps, λ=7.5 bps, 2,500 steps

Metric	Value
`final_pnl`	+1.725
`sharpe_annualized`	+2.314
`max_drawdown`	−9.472
`fills`	377
`fees_paid`	−1.165 (rebate)
`avg_abs_inventory`	2.65
`inventory_half_life`	118 steps
`realized_spread_avg`	0.003
`adverse_selection_avg`	0.001

Diagnostics on synthetic data – not a return claim.

Scope. Everything above is run against a synthetic GBM+regime-switch generator. The numbers are intended to validate that the measurement apparatus is correct and the strategy logic responds to frictions the way theory predicts – not to make return claims.

Engineering posture. Typed dataclasses, clean `src/` / `scripts/` / `notebooks/` boundaries, 24 tests across engine guards, depth, metrics, and behavioural properties, CI on Python 3.10–3.12 with papermill notebook smoke checks, deterministic seeds, every output reproducible from a JSON config.

3. Near-term roadmap

(i) L2/L3 order-book replay with queue-position modelling; (ii) replace synthetic series with Binance perpetual funding/basis history via public REST; (iii) exchange-specific funding intervals and borrow costs for realistic basis calibration; (iv) independent stochastic leg fills for the two-venue arb; (v) parameter-sweep tracking for multi-run aggregation.

Source: `github.com/<user>/microstructure-lab` • 18-slide Beamer deck available under `slides/`.